Evaluation (Part 1)

Dr. Mohammed Brahimi and Dr. Aicha Boutorh

- The most important part of the design of an evaluation experiment for a predictive model is ensuring that the data used to evaluate the model is not the same as the data used to train the model.

- The purpose of evaluation is threefold:
  1. to determine which model is the most suitable for a task
  2. to estimate how the model will perform
  3. to convince users that the model will meet their needs

Standard Approach: Measuring Misclassification Rate on a Hold-out Test Set

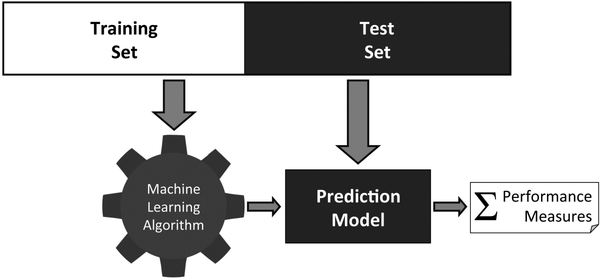


Figure: The process of building and evaluating a model using a **hold-out test set**.

Table: A sample test set with model predictions.

ID	Target	Pred.	Outcome	ID	Target	Pred.	Outcome
1	spam	ham	FN	11	ham	ham	TN
2	spam	ham	FN	12	spam	ham	FN
3	ham	ham	TN	13	ham	ham	TN
4	spam	spam	TP	14	ham	ham	TN
5	ham	ham	TN	15	ham	ham	TN
6	spam	spam	TP	16	ham	ham	TN
7	ham	ham	TN	17	ham	spam	FP
8	spam	spam	TP	18	spam	spam	TP
9	spam	spam	TP	19	ham	ham	TN
10	spam	spam	TP	20	ham	spam	FP

$$\text{misclassification rate} = \frac{\text{number incorrect predictions}}{\text{total predictions}} \tag{1}$$

$$\text{misclassification rate} = \frac{\text{number incorrect predictions}}{\text{total predictions}} \tag{1}$$

$$\text{misclassification rate} = \frac{(2+3)}{(6+9+2+3)} = 0.25$$

- For binary prediction problems there are 4 possible outcomes:
  1. True Positive (TP)
  2. True Negative (TN)
  3. False Positive (FP)
  4. False Negative (FN)

Table: The structure of a confusion matrix.

		Prediction	
		positive	**negative**
Target	**positive**	*TP*	*FN*
	negative	*FP*	*TN*

Table: A confusion matrix for the set of predictions shown in Table 1 [7].

		Prediction	
		'spam'	***'ham'***
Target	***'spam'***	6	3
	'ham'	2	9

$$\text{misclassification accuracy} = \frac{(FP + FN)}{(TP + TN + FP + FN)} \quad (2)$$

$$\text{misclassification accuracy} = \frac{(FP + FN)}{(TP + TN + FP + FN)} \quad (2)$$

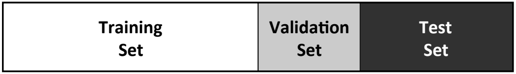
$$\text{misclassification accuracy} = \frac{(2 + 3)}{(6 + 9 + 2 + 3)} = 0.25$$

$$\text{classification accuracy} = \frac{(TP + TN)}{(TP + TN + FP + FN)} \quad (3)$$

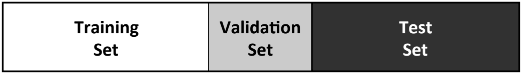
$$\text{classification accuracy} = \frac{(TP + TN)}{(TP + TN + FP + FN)} \tag{3}$$

$$\text{classification accuracy} = \frac{(6 + 9)}{(6 + 9 + 2 + 3)} = 0.75$$

Designing Evaluation Experiments



(a) A 50:20:30 split



(b) A 40:20:40 split

Figure: Hold-out sampling can divide the full data into training, validation, and test sets.

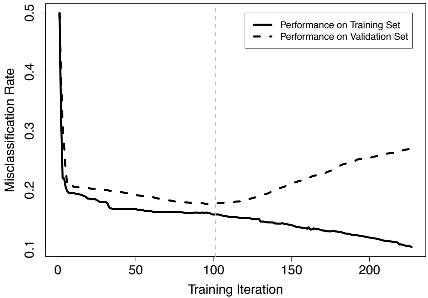


Figure: Using a validation set to avoid overfitting in iterative machine learning algorithms.

Fold	Confusion Matrix				Class Accuracy
			Prediction		
			'lateral'	*'frontal'*	
1	Target	*'lateral'*	43	9	81%
		'frontal'	10	38	
			Prediction		
			'lateral'	*'frontal'*	
2	Target	*'lateral'*	46	9	88%
		'frontal'	3	42	
			Prediction		
			'lateral'	*'frontal'*	
3	Target	*'lateral'*	51	10	82%
		'frontal'	8	31	
			Prediction		
			'lateral'	*'frontal'*	
4	Target	*'lateral'*	51	8	85%
		'frontal'	7	34	
			Prediction		
			'lateral'	*'frontal'*	
5	Target	*'lateral'*	46	9	84%
		'frontal'	7	38	
			Prediction		
			'lateral'	*'frontal'*	
Overall	Target	*'lateral'*	237	45	84%
		'frontal'	35	183	

k-Fold Cross Validation

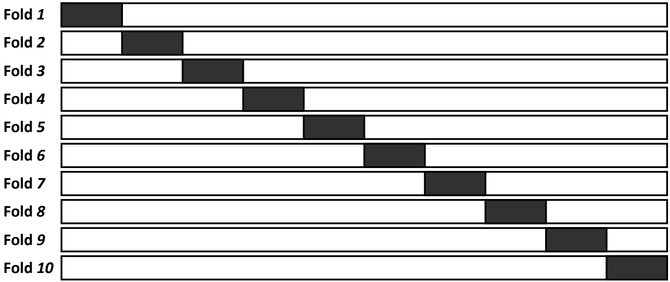


Figure: The division of data during the ***k*-fold cross validation** process. Black rectangles indicate test data, and white spaces indicate training data.

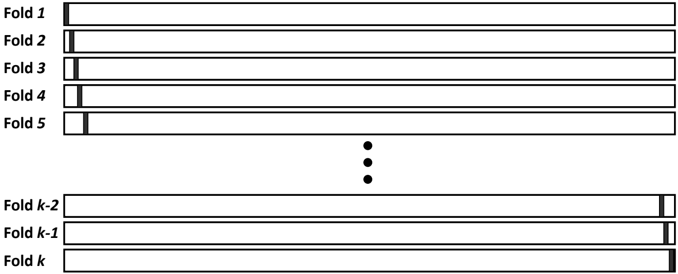


Figure: The division of data during the **leave-one-out cross validation** process. Black rectangles indicate instances in the test set, and white spaces indicate training data.

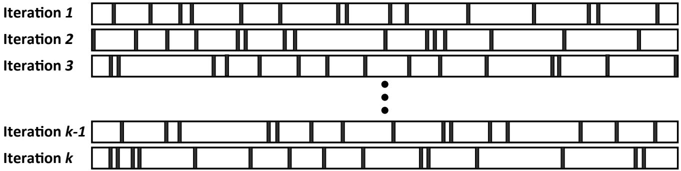


Figure: The division of data during the $\epsilon 0$ bootstrap process. Black rectangles indicate test data, and white spaces indicate training data.

Out-of-time Sampling

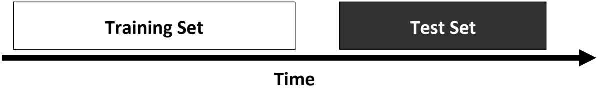


Figure: The **out-of-time sampling** process.

Performance Measures: Categorical Targets

$$\text{TPR} = \frac{TP}{(TP + FN)} \tag{4}$$

$$\text{TNR} = \frac{TN}{(TN + FP)} \tag{5}$$

$$\text{FPR} = \frac{FP}{(TN + FP)} \tag{6}$$

$$\text{FNR} = \frac{FN}{(TP + FN)} \tag{7}$$

$$
\begin{aligned}
\text{TPR} &= \frac{6}{(6+3)} &= 0.667\\
\text{TNR} &= \frac{9}{(9+2)} &= 0.818\\
\text{FPR} &= \frac{2}{(9+2)} &= 0.182\\
\text{FNR} &= \frac{3}{(6+3)} &= 0.333
\end{aligned}
$$

Precision, Recall and F_1 Measure

$$\text{precision} = \frac{TP}{(TP + FP)} \tag{8}$$

$$\text{recall} = \frac{TP}{(TP + FN)} \tag{9}$$

$$
\begin{aligned}
\text{precision} &= \frac{6}{(6+2)} = 0.75\\
\text{recall} &= \frac{6}{(6+3)} = 0.667
\end{aligned}
$$

$$\text{F}_1\text{-measure} = 2 \times \frac{(\text{precision} \times \text{recall})}{(\text{precision} + \text{recall})} \tag{10}$$

$$F_1\text{-measure} = 2 \times \frac{(\text{precision} \times \text{recall})}{(\text{precision} + \text{recall})} \tag{10}$$

$$\begin{aligned} F_1\text{-measure} &= 2 \times \frac{\left(\frac{6}{(6+2)} \times \frac{6}{(6+3)}\right)}{\left(\frac{6}{(6+2)} + \frac{6}{(6+3)}\right)} \\ &= 0.706 \end{aligned}$$

Table: A confusion matrix for a *k*-NN model trained on a churn prediction problem.

		Prediction	
		'non-churn'	*'churn'*
Target	*'non-churn'*	90	0
	'churn'	9	1

Table: A confusion matrix for a naive Bayes model trained on a churn prediction problem.

		Prediction	
		'non-churn'	*'churn'*
Target	*'non-churn'*	70	20
	'churn'	2	8

$$\text{average class accuracy} = \frac{1}{|levels(t)|} \sum_{l \in levels(t)} \text{recall}_l \qquad (11)$$

Average Class Accuracy

$$\text{average class accuracy}_{\text{HM}} = \frac{1}{\frac{1}{|\mathit{levels}(t)|} \sum_{l \in \mathit{levels}(t)} \frac{1}{\text{recall}_l}} \tag{12}$$

$$\frac{1}{\frac{1}{2}\left(\frac{1}{1.0}+\frac{1}{0.1}\right)}=\frac{1}{5.5}=18.2\%$$

$$\frac{1}{\frac{1}{2}\left(\frac{1}{0.778}+\frac{1}{0.800}\right)}=\frac{1}{1.268}=78.873\%$$

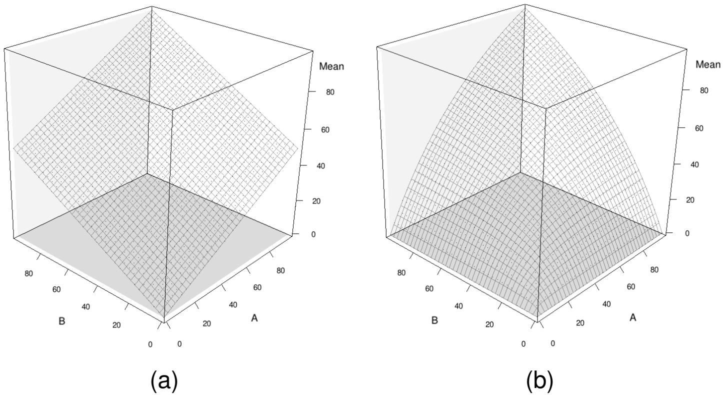


Figure: Surfaces generated by calculating (a) the **arithmetic mean** and (b) the **harmonic mean** of all combinations of features A and B that range from 0 to 100.

- It is not always correct to treat all outcomes equally
- In these cases, it is useful to take into account the cost of the different outcomes when evaluating models

Table: The structure of a **profit matrix**.

		Prediction positive	Prediction negative
Target	**positive**	TP_{Profit}	FN_{Profit}
	negative	FP_{Profit}	TN_{Profit}

Table: The **profit matrix** for the pay-day loan credit scoring problem.

		Prediction	
		'good'	***'bad'***
Target	***'good'***	140	-140
	'bad'	-700	0

Table: (a) The confusion matrix for a *k*-NN model trained on the pay-day loan credit scoring problem (average class accuracy$_{HM}$ $= 83.824\%$); (b) the confusion matrix for a decision tree model trained on the pay-day loan credit scoring problem (average class accuracy$_{HM}$ $= 80.761\%$).

(a) *k*-NN model

		Prediction	
		'good'	***'bad'***
Target	***'good'***	57	3
	'bad'	10	30

(b) decision tree

		Prediction	
		'good'	***'bad'***
Target	***'good'***	43	17
	'bad'	3	37

Table: (a) Overall profit for the *k*-NN model using the profit matrix in Table 7 [39] and the **confusion matrix** in Table 8(a) [40]; (b) overall profit for the decision tree model using the profit matrix in Table 7 [39] and the **confusion matrix** in Table 8(b) [40].

(a) *k*-NN model

		Prediction	
		'good'	*'bad'*
Target	*'good'*	7 980	−420
	'bad'	−7 000	0
	Profit		**560**

(b) decision tree

		Prediction	
		'good'	*'bad'*
Target	*'good'*	6 020	−2 380
	'bad'	−2 100	0
	Profit		**1 540**

Summary

Slide Acknowledgment

The slides used in this course are based on the official textbook materials, with modifications made where necessary to suit the course requirements and enhance the learning experience.